8.0.1 ISRO CSE 2008 | Question: 48

Which of the following is termed as minimum error code ?

A. Binary code B. Gray code C. Excess-3 code D. Octal code

isro2008 digital-logic binary-codes

Answer key

8.0.2 ISRO CSE 2014 | Question: 26

The output of a tristate buffer when the enable input in 0 is

A. Always 0
B. Always 1
C. Retains the last value when enable input was high
D. Disconnected state

isro2014 digital-logic translation-lookaside-buffer

Answer key

8.0.3 ISRO CSE 2016 | Question: 15

The Excess-3 code is also called

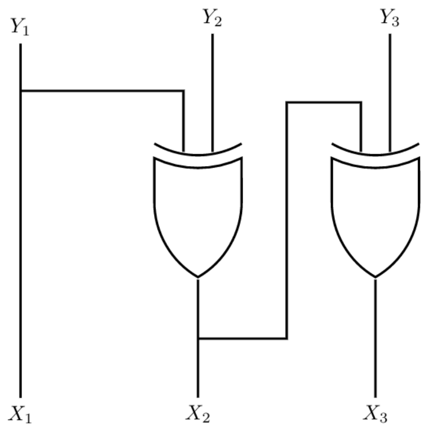
A. Cyclic Redundancy Code
B. Weighted Code
C. Self-Complementing Code
D. Algebraic Code

isro2016 digital-logic binary-codes

Answer key

8.0.4 ISRO CSE 2016 | Question: 12

The logic circuit given below converts a binary code $Y_1Y_2Y_3$ into



A. Excess-3 code B. Gray code C. BCD code D. Hamming code

digital-logic binary-codes isro2016

Answer key

8.0.5 ISRO CSE 2023 | Question: 74

A truth table will need how many rows if there are n variables.

A. $2^{*}n$ B. n C. 2^{n} D. None of the above

isro-cse-2023 digital-logic easy

Answer key

8.0.6 ISRO CSE 2023 | Question: 10

A logic circuit that provides a LOW output when both inputs are HIGH or both inputs are LOW is

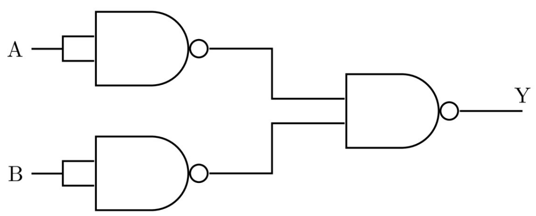
A. AND B. NAND C. XNOR D. XOR

isro-cse-2023 digital-logic easy

Answer key

8.0.7 ISRO CSE 2023 | Question: 9

The logical operation of the following circuit is



A. XOR B. NAND C. AND D. OR

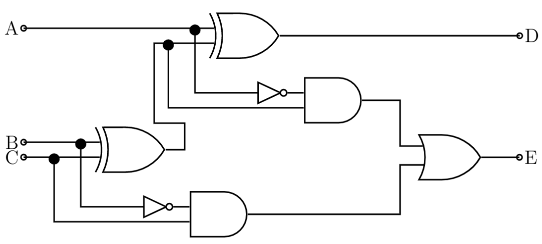
isro-cse-2023 digital-logic compiler-design

Answer key

8.1 Adder (6)

8.1.1 Adder: ISRO CSE 2007 | Question: 03

The circuit shown in the given figure is a



A. full adder
B. full subtracter
C. shift register
D. decade counter

isro2007 digital-logic digital-circuits adder

Answer key

8.1.2 Adder: ISRO CSE 2016 | Question: 9

For a binary half-subtractor having two inputs A and B, the correct set of logical outputs

D $(= \mathrm{A\ minus\ B})$ and $\mathrm{X}(= \mathrm{borrow})$ are

A. $\mathrm{D} = \mathrm{AB} + \overline{A}\mathrm{B}, \mathrm{X} = \overline{\mathrm{A}}\mathrm{B}$
B. $\mathrm{D} = \overline{A}\mathrm{B} + A\overline{\mathrm{B}}, \mathrm{X} = \mathrm{A}\overline{\mathrm{B}}$
C. $\mathrm{D} = \overline{A}\mathrm{B} + A\overline{\mathrm{B}}, \mathrm{X} = \overline{\mathrm{A}}\mathrm{B}$
D. $\mathrm{D} = \mathrm{AB} + \overline{A}\mathrm{B}, \mathrm{X} = \mathrm{A}\overline{\mathrm{B}}$

isro2016 digital-logic adder

Answer key

8.1.3 Adder: ISRO CSE 2017 | Question: 24

When two n-bit binary numbers are added the sum will contain at the most

A. n bits
B. $n+2$ bits
C. $n+3$ bits
D. $n+1$ bits

isro2017 digital-logic adder

Answer key

8.1.4 Adder: ISRO CSE 2020 | Question: 9

In a 8-bit ripple carry adder using identical full adders, each full adder takes 34 ns for computing sum. If the time taken for 8-bit addition is 90 ns, find time taken by each full adder to find carry.

A. 6 ns
B. 7 ns
C. 10 ns
D. 8 ns

isro-2020 digital-logic combinational-circuit adder normal

Answer key

8.1.5 Adder: ISRO-DEC2017-71

A 32-bit adder is formed by cascading 4-bit CLA adder.The gate delays (latency) for getting the sum bits is

A. 16
B. 18
C. 17
D. 19

isrodec2017 co-and-architecture digital-circuits adder

Answer key

8.1.6 Adder: ISRO2015-7

If half adders and full adders are implements using gates, then for the addition of two 17 bit numbers (using minimum gates) the number of half adders and full adders required will be

A. $0, 17$
B. $16, 1$
C. $1, 16$
D. $8, 8$

isro2015 digital-logic adder

Answer key

8.2 Bcd (1)

8.2.1 Bcd: ISRO-2013-11

When two BCD numbers 0×14 and 0×08 are added what is the binary representation of the resultant number ?

A. 0×22
B. $0 \times 1c$
C. 0×16
D. results in overflow

isro2013 number-representation bcd

Answer key

8.3 Boolean Algebra (13)

8.3.1 Boolean Algebra: ISRO CSE 2007 | Question: 01

The Boolean expression $Y = (A + \overline{B} + \overline{A}B)\overline{C}$ is given by

A. $A\overline{C}$
B. $B\overline{C}$
C. $\overline{C}$
D. AB

isro2007 digital-logic boolean-algebra

Answer key

8.3.2 Boolean Algebra: ISRO CSE 2008 | Question: 11

The Boolean theorem $AB + \overline{A}C + BC = AB + \overline{A}C$ corresponds to

A. $(A+B) \bullet (\overline{A} + C) \bullet (B+C) = (A+B) \bullet (\overline{A} + C)$
B. $AB + \overline{A}C + BC = AB + BC$
C. $AB + \overline{A}C + BC = (A+B) \bullet (\overline{A} + C) \bullet (B+C)$
D. $(A+B) \bullet (\overline{A} + C) \bullet (B+C) = AB + \overline{A}C$

isro2008 digital-logic boolean-algebra

Answer key

8.3.3 Boolean Algebra: ISRO CSE 2008 | Question: 24

The Boolean expression $(A + \overline{C})(\overline{B} + \overline{C})$ simplifies to

A. $\overline{C} + A\overline{B}$
B. $\overline{C}(\overline{A} + B)$
C. $\overline{B}\,\overline{C} + A\overline{B}$
D. None of these

isro2008 digital-logic boolean-algebra

Answer key

8.3.4 Boolean Algebra: ISRO CSE 2008 | Question: 25

In the expression $\overline{A}(\overline{A} + \overline{B})$ by writing the first term A as $A + 0$, the expression is best simplified as

A. $A+AB$
B. AB
C. A
D. $A+B$

isro2008 digital-logic boolean-algebra

Answer key

8.3.5 Boolean Algebra: ISRO CSE 2008 | Question: 28

Which of the following is not a valid rule of XOR?

A. $0 \text{ XOR } 0 = 0$
B. $1 \text{ XOR } 1 = 1$
C. $1 \text{ XOR } 0 = 1$
D. $\text{B XOR B} = 0$

isro2008 digital-logic boolean-algebra

Answer key☟

8.3.6 Boolean Algebra: ISRO CSE 2009 | Question: 20

Consider the following boolean function of four variables $f(w,x,y,z) = \Sigma(1,3,4,6,9,11,12,14)$, the function is

A. Independent of one variable
B. Independent of two variables
C. Independent of three variables
D. Dependent on all variables

isro2009 digital-logic boolean-algebra

Answer key☟

8.3.7 Boolean Algebra: ISRO CSE 2011 | Question: 30

In Boolean algebra, rule $(\mathrm{X}+\mathrm{Y})(\mathrm{X}+\mathrm{Z}) =$

A. $\mathrm{Y}+\mathrm{XZ}$
B. $\mathrm{X}+\mathrm{YZ}$
C. $\mathrm{XY}+\mathrm{Z}$
D. $\mathrm{XZ}+\mathrm{Y}$

isro2011 digital-logic boolean-algebra

Answer key☟

8.3.8 Boolean Algebra: ISRO CSE 2011 | Question: 6

Evaluate $(\mathrm{X} \text{ xor } \mathrm{Y}) \text{ xor } \mathrm{Y}$?

A. All 1's
B. All 0's
C. X
D. Y

isro2011 digital-logic boolean-algebra

Answer key☟

8.3.9 Boolean Algebra: ISRO CSE 2014 | Question: 56

Which of the following is not valid Boolean algebra rule?

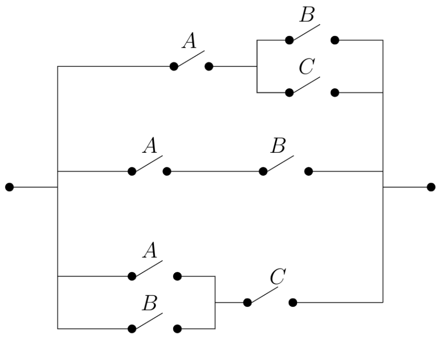
A. $\mathrm{X.X} = \mathrm{X}$
B. $(\mathrm{X}+\mathrm{Y}).\mathrm{X} = \mathrm{X}$
C. $\overline{X}+\mathrm{XY} = \mathrm{Y}$
D. $(\mathrm{X}+\mathrm{Y}).(\mathrm{X}+\mathrm{Z}) = \mathrm{X}+\mathrm{YZ}$

digital-logic boolean-algebra isro2014

Answer key☟

8.3.10 Boolean Algebra: ISRO CSE 2016 | Question: 8

The minimum Boolean expression for the following circuit is



A. $\mathrm{AB}+\mathrm{AC}+\mathrm{BC}$
B. $\mathrm{A}+\mathrm{BC}$
C. $\mathrm{A}+\mathrm{B}$
D. $\mathrm{A}+\mathrm{B}+\mathrm{C}$

isro2016 digital-logic boolean-algebra

Answer key

8.3.11 Boolean Algebra: ISRO CSE 2018 | Question: 62

Any set of Boolean operation that is sufficient to represent all Boolean expression is said to be complete. Which of the following is not complete ?

A. $\{\mathrm{AND}, \mathrm{OR}\}$
B. $\{\mathrm{AND}, \mathrm{NOT}\}$
C. $\{\mathrm{NOT}, \mathrm{OR}\}$
D. $\{\mathrm{NOR}\}$

isro2018 digital-logic boolean-algebra

Answer key

8.3.12 Boolean Algebra: ISRO2015-10

The boolean expression $\mathrm{AB}+\mathrm{AB}'+\mathrm{A}'\mathrm{C}+\mathrm{AC}$ is independent of the boolean variable

A. A
B. B
C. C
D. None of these

isro2015 digital-logic boolean-algebra

Answer key

8.3.13 Boolean Algebra: ISRO2015-5

The complement of the Boolean expression $\mathrm{AB}(\overline{\mathrm{B}}\mathrm{C}+\mathrm{AC})$ is

A. $(\overline{\mathrm{A}}+\overline{\mathrm{B}})+(\mathrm{B}+\overline{\mathrm{C}}).(\overline{\mathrm{A}}+\overline{\mathrm{C}})$
B. $(\overline{\mathrm{A}}.\overline{\mathrm{B}})+(\mathrm{B}\overline{\mathrm{C}}+\overline{\mathrm{A}}\ \overline{\mathrm{C}})$
C. $(\overline{\mathrm{A}}+\overline{\mathrm{B}}).(\mathrm{B}+\overline{\mathrm{C}})+(\mathrm{A}+\overline{\mathrm{C}})$
D. $(\mathrm{A}+\mathrm{B}).(\overline{\mathrm{B}}+\mathrm{C})\ (\mathrm{A}+\mathrm{C})$

isro2015 digital-logic boolean-algebra

Answer key

8.4 Booths Algorithm (1)

8.4.1 Booths Algorithm: ISRO CSE 2009 | Question: 40 , GATE2008-IT_42

The two numbers given below are multiplied using the Booth's algorithm

Multiplicand : $0101\ 1010\ 1110\ 1110$

Multiplier : $0111\ 0111\ 1011\ 1101$

How many additions/subtractions are required for the multiplication of the above two numbers?

A. 6 B. 8 C. 10 D. 12

isro2009 digital-logic booths-algorithm

Answer key

8.5 CO and Architecture (2)

8.5.1 CO and Architecture: ISRO CSE 2023 | Question: 25

The operation executed on data stored in registers is called

A. Macro-operation
B. Micro-operation
C. Bit-operation
D. Byte-operation

isro-cse-2023 co-and-architecture

Answer key

8.5.2 CO and Architecture: ISRO CSE 2023 | Question: 8

A full adder circuit requires

A. two inputs and two outputs
B. two inputs and three outputs
C. three inputs and two outputs
D. three inputs and one output

isro-cse-2023 digital-logic co-and-architecture

Answer key

8.6 Canonical Normal Form (2)

8.6.1 Canonical Normal Form: ISRO CSE 2016 | Question: 16

The simplified SOP (Sum of Product) from the Boolean expression

$$(\mathrm{P}+\overline{\mathrm{Q}}+\overline{\mathrm{R}}).(\mathrm{P}+\mathrm{Q}+\mathrm{R})\ .\ (\mathrm{P}+\mathrm{Q}+\overline{\mathrm{R}})$$

is

A. $(\overline{\mathrm{P}}.\mathrm{Q}+\overline{\mathrm{R}})$
B. $(\mathrm{P}+\mathrm{Q}.\overline{\mathrm{R}})$
C. $(\mathrm{P}.\overline{\mathrm{Q}}+\mathrm{R})$
D. $(\mathrm{P}.\mathrm{Q}+\mathrm{R})$

digital-logic canonical-normal-form isro2016

Answer key

8.6.2 Canonical Normal Form: ISRO-2013-28

The most simplified form of the Boolean function
$x(A,B,C,D)=\sum(7,8,9,10,11,12,13,14,15)$
(expressed in sum of minterms) is?

A. A + A'BCD B. AB + CD C. A + BCD D. ABC + D

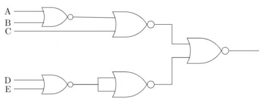
isro2013 digital-logic canonical-normal-form

Answer key

8.7 Circuit Output (14)

8.7.1 Circuit Output: ISRO CSE 2007 | Question: 02

The circuit shown in the following figure realizes the function.



A. $(\overline{A+B}+C)(\overline{D}\,\overline{E})$

B. $(\overline{A+B}+C)(D\overline{E})$

C. $(A+\overline{B+C})(\overline{D}E)$

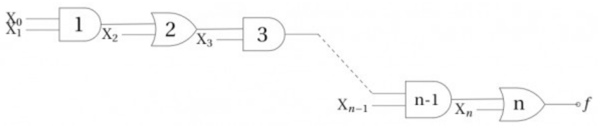
D. $(A+B+\overline{C})(\overline{D}\,\overline{E})$

isro2007 digital-logic digital-circuits circuit-output

Answer key

8.7.2 Circuit Output: ISRO CSE 2008 | Question: 12

In the given network of AND and OR gates f can be written as



A. $X_0X_1X_2\dots X_n + X_1X_2\dots X_n + X_2X_3\dots X_n + \dots + X_n$

B. $X_0X_1 + X_2X_3 + \dots X_{n-1}X_n$

C. $X_0 + X_1 + X_2 + \dots + X_n$

D. $X_0X_1 + X_3\dots X_{n-1} + X_2X_3 + X_5\dots X_{n-1} + \dots + X_{n-2}X_{n-1} + X_n$

isro2008 digital-logic circuit-output

Answer key

8.7.3 Circuit Output: ISRO CSE 2008 | Question: 26

The logic operations of two combinational circuits in Figure-I and Figure -II are

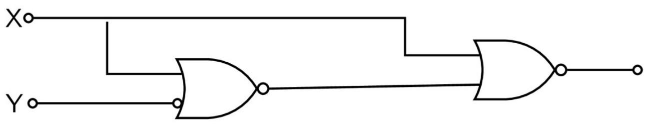


Figure -I

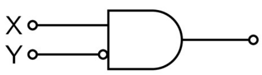


Figure -II

A. entirely different

B. identical

C. complementary

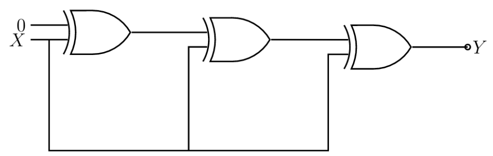
D. dual

isro2008 digital-logic circuit-output

Answer key

8.7.4 Circuit Output: ISRO CSE 2008 | Question: 27

The output Y of the given circuit



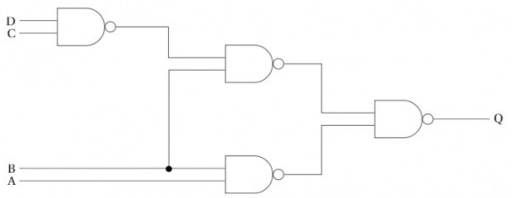
A. 1 B. 0 C. X D. X'

isro2008 digital-logic circuit-output

Answer key☟

8.7.5 Circuit Output: ISRO CSE 2014 | Question: 15

Consider the logic circuit given below:



$Q =$ ___________?

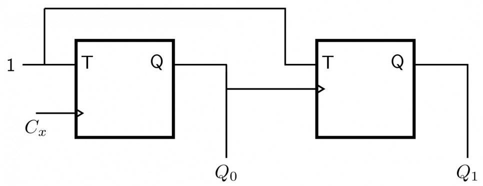
A. $\overline{A}C + B\overline{C} + CD$
B. $ABC + \overline{C}D$
C. $AB + B\overline{C} + B\overline{D}$
D. $A\overline{B} + A\overline{C} + \overline{C}D$

isro2014 digital-logic circuit-output

Answer key☟

8.7.6 Circuit Output: ISRO CSE 2014 | Question: 21, UGCNET-Dec2012-III: 23, UGCNET-Dec2013-III: 22

What are the final values of Q_1 and Q_0 after 4 clock cycles, if initial values are 00 in the sequential circuit shown below:



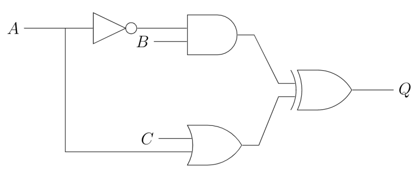
A. 11 B. 01 C. 10 D. 00

isro2014 digital-logic circuit-output ugcnetcse-dec2012-paper3 ugcnetcse-dec2013-paper3

Answer key☟

8.7.7 Circuit Output: ISRO CSE 2014 | Question: 53

Consider the logic circuit given below.



The inverter, AND and OR gates have delays of $6, 10$ and 11 nanoseconds respectively. Assuming that wire delays are negligible, what is the duration of glitch for Q before it becomes stable?

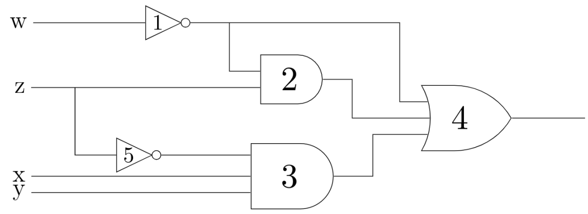
A. 5 B. 11 C. 16 D. 27

isro2014 digital-logic circuit-output

Answer key

8.7.8 Circuit Output: ISRO CSE 2016 | Question: 10

Consider the following gate network



Which one of the following gates is redundant?

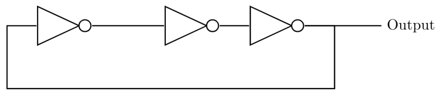
A. Gate No. 1 B. Gate No. 2 C. Gate No. 3 D. Gate No. 4

isro2016 digital-logic circuit-output

Answer key

8.7.9 Circuit Output: ISRO CSE 2016 | Question: 13

The circuit given in the figure below is

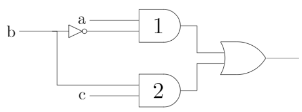


A. An oscillating circuit and its output is square wave
B. The one whose output remains stable in '1' state
C. The one having output remains stable in '0' state
D. has a single pulse of three times propagation delay

isro2016 digital-logic circuit-output

Answer key

8.7.10 Circuit Output: ISRO CSE 2018 | Question: 9



In the diagram above, the inverter (NOT gate) and the AND-gates labeled 1 and 2 have delays of $9, 10$ and 12 nanoseconds (ns), respectively. Wire delays are negligible. For certain values a and c, together with certain transition of b, a glitch (spurious output) is generated for a short time, after which the output assumes its correct value. The duration of glitch is:

a. 7 ns b. 9 ns c. 11 ns d. 13 ns

isro2018 digital-logic circuit-output

Answer key

8.7.11 Circuit Output: ISRO CSE 2020 | Question: 11

Minimum number of NAND gates required to implement the following binary equation

$Y = (\overline{A} + \overline{B})(C + D)$

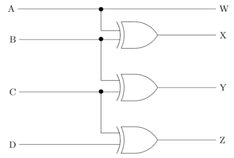
A. 4 B. 5 C. 3 D. 6

isro-2020 digital-logic combinational-circuit circuit-output normal

Answer key

8.7.12 Circuit Output: ISRO CSE 2020 | Question: 12

If $ABCD$ is a 4-bit binary number, then what is the code generated by the following circuit?



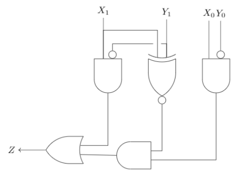
A. BCD code B. Gray code C. 8421 code D. Excess-3 code

isro-2020 digital-logic combinational-circuit circuit-output normal

Answer key

8.7.13 Circuit Output: ISRO CSE 2020 | Question: 66

The following circuit compares two 2-bit binary numbers, X and Y represented by X_1X_0 and Y_1Y_0 respectively. (X_0 and Y_0 represent Least Significant Bits)



Under what conditions Z will be 1?

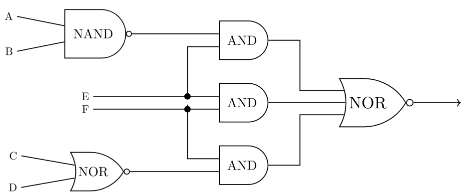
A. $X > Y$ B. $X < Y$ C. $X = Y$ D. $X! = Y$

isro-2020 digital-logic digital-circuits circuit-output normal

Answer key

8.7.14 Circuit Output: ISRO CSE 2020 | Question: 77

Consider the following circuit



The function by the network above is

A. $\overline{AB}E + EF + \overline{CD}F$
B. $(\overline{E} + AB\overline{F})(C + D + \overline{F})$
C. $(\overline{AB} + E)(\overline{E} + \overline{F})(C + D + \overline{F})$
D. $(A + B)\overline{E} + \overline{EF} + CD\overline{F}$

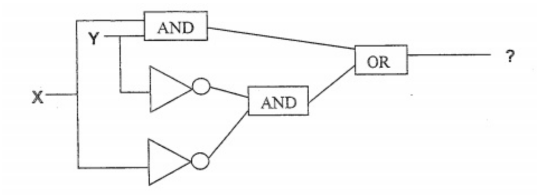
isro-2020 digital-logic combinational-circuit circuit-output normal

Answer key

8.8 Combinational Circuit (1)

8.8.1 Combinational Circuit: ISRO CSE 2011 | Question: 47

The output expression of the following gate network is



A. $\mathrm{X.Y} + \overline{\mathrm{X}}.\,\overline{\mathrm{Y}}$
B. $\mathrm{X.Y} + \mathrm{X.Y}$
C. $\mathrm{X.Y}$
D. $\mathrm{X} + \mathrm{Y}$

isro2011 digital-logic combinational-circuit digital-circuits

Answer key

8.9 Decoder (1)

8.9.1 Decoder: GATE CSE 2007 | Question: 8, ISRO2011-31

How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

A. 7 B. 8 C. 9 D. 10

gatecse-2007 digital-logic normal isro2011 decoder

Answer key

8.10 Digital Circuits (2)

8.10.1 Digital Circuits: ISRO CSE 2017 | Question: 26

Advantage of synchronous sequential circuits over asynchronous one is :

A. Lower hardware requirement
B. Better noise immunity
C. Faster operation
D. All of the above

isro2017 digital-logic digital-circuits bad-question

Answer key

8.10.2 Digital Circuits: ISRO-DEC2017-72

We consider the addition of two $2's$ compliment numbers $b_{n-1}b_{n-2}\dots b_0$ and $a_{n-1}a_{n-2}\dots a_0$. A binary adder for adding two unsigned binary numbers is used to add two binary numbers.The sum is denoted by $c_{n-1}c_{n-2}\dots c_0$.The carry out is denoted by c_{out}.The overflow condition is identified by

A. $c_{out}\left(\overline{a_{n-1} \oplus b_{n-1}}\right)$

B. $\overline{a_{n-1}}b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}b_{n-1}}c_{n-1}$

C. $c_{out} \oplus c_{n-1}$

D. $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

isrodec2017 digital-circuits number-representation isro

Answer key☟

8.11 Digital Counter (2)

8.11.1 Digital Counter: ISRO CSE 2007 | Question: 34

Ring counter is analogous to

A. Toggle Switch
B. Latch
C. Stepping Switch
D. S-R flip flop

isro2007 digital-logic digital-counter

Answer key☟

8.11.2 Digital Counter: ISRO2015-4

A modulus -12 ring counter requires a minimum of

A. 10 flip-flops
B. 12 flip-flops
C. 8 flip-flops
D. 6 flip-flops

isro2015 digital-logic digital-counter

Answer key☟

8.12 Flip Flop (6)

8.12.1 Flip Flop: GATE CSE 2004 | Question: 18, ISRO2007-31

In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in

A. $Q = 0, Q' = 1$
B. $Q = 1, Q' = 0$
C. $Q = 1, Q' = 1$
D. Indeterminate states

gatecse-2004 digital-logic easy isro2007 flip-flop

Answer key☟

8.12.2 Flip Flop: ISRO CSE 2007 | Question: 05

The characteristic equation of an SR flip-flop is given by :

A. $Q_{n+1} = S + RQ_n$
B. $Q_{n+1} = R\bar{Q}_n + \bar{S}Q_n$
C. $Q_{n+1} = \bar{S} + RQ_n$
D. $Q_{n+1} = S + \bar{R}Q_n$

isro2007 digital-logic flip-flop

Answer key☟

8.12.3 Flip Flop: ISRO CSE 2011 | Question: 74

In an RS flip-flop, if the S line (Set line) is set high (1) and the R line (Reset line) is set low (0), then the state of the flip-flop is :

A. Set to 1
B. Set to 0
C. No change in state
D. Forbidden

isro2011 digital-logic flip-flop

Answer key☟

8.12.4 Flip Flop: ISRO CSE 2016 | Question: 18

The functional difference between SR flip-flop and J-K flip-flop is that :

A. J-K flip-flop is faster than SR flip-flop
B. J-K flip-flop has a feedback path
C. J-K flip-flop accepts both inputs 1
D. None of them

digital-logic flip-flop isro2016

Answer key☟

8.12.5 Flip Flop: ISRO CSE 2020 | Question: 80

A new flipflop with inputs X and Y, has the following property

X	Y	Current state	Next state
0	0	Q	1
0	1	Q	$\overline{Q}$
1	1	Q	0
1	0	Q	Q

Which of the following expresses the next state in terms of $X, Y,$ current state?

A. $(\overline{X} \wedge \overline{Q}) \vee (\overline{Y} \wedge Q)$
B. $(\overline{X} \wedge Q) \vee (\overline{Y} \wedge \overline{Q})$
C. $(X \wedge \overline{Q}) \vee (Y \wedge Q)$
D. $(X \wedge \overline{Q}) \vee (\overline{Y} \wedge Q)$

isro-2020 digital-logic sequential-circuit flip-flop normal

Answer key☟

8.12.6 Flip Flop: ISRO-2013-30

In a three stage counter, using RS flip flops what will be the value of the counter after giving 9 pulses to its input ? Assume that the value of counter before giving any pulses is 1 :

A. 1
B. 2
C. 9
D. 10

isro2013 digital-logic flip-flop

Answer key☟

8.13 Floating Point Representation (6)

8.13.1 Floating Point Representation: ISRO CSE 2007 | Question: 30

0.75 decimal system is equivalent to ____ in octal system

A. 0.60
B. 0.52
C. 0.54
D. 0.50

isro2007 digital-logic number-representation floating-point-representation

Answer key☟

8.13.2 Floating Point Representation: ISRO CSE 2007 | Question: 36

Consider a computer system that stores a floating-point numbers with 16-bit mantissa and an 8-bit exponent, each in two's complement. The smallest and largest positive values which can be stored are :

A. 1×10^{-128} and $2^{15} \times 10^{128}$
B. 1×10^{-256} and $2^{15} \times 10^{255}$
C. 1×10^{-128} and $2^{15} \times 10^{127}$
D. 1×10^{-128} and $2^{15} - 1 \times 10^{127}$

isro2007 digital-logic number-representation floating-point-representation

Answer key

8.13.3 Floating Point Representation: ISRO CSE 2008 | Question: 23

A computer uses 8 digit mantissa and 2 digit exponent. If $a = 0.052$ and $b = 28E + 11$ then $b + a - b$ will :

A. result in an overflow error
B. result in an underflow error
C. be 0
D. be $5.28E$+11

isro2008 digital-logic number-representation floating-point-representation

Answer key

8.13.4 Floating Point Representation: ISRO CSE 2011 | Question: 8

What is the decimal value of the floating-point number $C1D00000$ (hexadecimal notation)? (Assume 32-bit, single precision floating point IEEE representation)

A. 28
B. -15
C. -26
D. -28

isro2011 digital-logic number-representation floating-point-representation ieee-representation

Answer key

8.13.5 Floating Point Representation: ISRO-DEC2017-75

In IEEE floating point representation, the hexadecimal number 0xC0000000 corresponds to

A. -3.0
B. -1.0
C. -4.0
D. -2.0

isrodec2017 ieee-representation floating-point-representation digital-logic

Answer key

8.13.6 Floating Point Representation: ISRO2015-1

Which of the given number has its IEEE-754 32-bit floating point representation as $(0\ 10000000\ 110\ 0000\ 0000\ 0000\ 0000\ 0000)$

A. 2.5
B. 3.0
C. 3.5
D. 4.5

isro2015 digital-logic number-representation floating-point-representation ieee-representation

Answer key

8.14 Functional Completeness (2)

8.14.1 Functional Completeness: ISRO-2013-22

Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is not complete ?

A. $\{NOT, OR\}$
B. $\{NOR\}$
C. $\{AND, OR\}$
D. $\{AND, NOT\}$

isro2013 functional-completeness digital-logic digital-circuits combinational-circuit

Answer key

8.14.2 Functional Completeness: ISRO-DEC2017-76

Which of the following set of components is sufficient to implement any arbitrary Boolean function?

A. XOR gates, NOT gates
B. AND gates, XOR gates and 1
C. 2 to 1 multiplexer
D. Three input gates that output $(A.B)+C$ for the inputs A, B, C

isrodec2017 functional-completeness digital-logic

Answer key

8.15 Hazards (1)

8.15.1 Hazards: ISRO CSE 2016 | Question: 11

The dynamic hazard problem occurs in

A. combinational circuit alone
B. sequential circuit only
C. Both (a) and (b)
D. None of the above

isro2016 digital-logic digital-circuits hazards

Answer key

8.16 IEEE Representation (1)

8.16.1 IEEE Representation: ISRO CSE 2014 | Question: 8

In the standard $\text{IEEE } 754$ single precision floating point representation, there is 1 bit for sign, 23 bits for fraction and 8 bits for exponent. What is the precision in terms of the number of decimal digits?

A. 5
B. 6
C. 7
D. 8

number-representation ieee-representation isro2014

Answer key

8.17 Interrupts (1)

8.17.1 Interrupts: ISRO CSE 2023 | Question: 5

In a vectored interrupt

A. The branch address is assigned to a fixed location in memory
B. The interrupting source supplies the branch information to the processor
C. The branch address is obtained from a register in the processor
D. None of the above

isro-cse-2023 co-and-architecture interrupts

Answer key☟

8.18 K Map (1)

8.18.1 K Map: ISRO CSE 2009 | Question: 19

The switching expression corresponding to $f(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}) = \Sigma(1,4,5,9,11,12)$ is

A. BC'D' + A'C'D + AB'D
B. ABC' + ACD + B'C'D
C. ACD' + A'BC' + AC'D'
D. A'BD + ACD' + BCD'

isro2009 digital-logic boolean-algebra k-map

Answer key☟

8.19 Logic (1)

8.19.1 Logic: ISRO CSE 2023 | Question: 29

The resulting logic circuit obtained when both inputs of NOR and NAND gates are connected together is:

A. AND
B. XOR
C. OR
D. NOT

isro-cse-2023 digital-logic logic circuit-output

Answer key☟

8.20 Matrix Chain Ordering (1)

8.20.1 Matrix Chain Ordering: ISRO CSE 2023 | Question: 72

The complexity of matrix multiplication of two matrices A and B whose orders are $\mathrm{m} \times \mathrm{n}$ and $\mathrm{n} \times \mathrm{p}$ respectively is

A. $\mathrm{O}(\mathrm{m} \times \mathrm{p})$
B. $O\left(m \times n^2 \times p\right)$
C. $\mathrm{O}\left(\mathrm{m} \times \mathrm{n} \times \mathrm{p}^2\right)$
D. $\mathrm{O}(\mathrm{m} \times \mathrm{n} \times \mathrm{p})$

isro-cse-2023 algorithms time-complexity matrix-chain-ordering

Answer key☟

8.21 Memory Interfacing (2)

8.21.1 Memory Interfacing: GATE CSE 2009 | Question: 7, ISRO2015-3

How many $32K \times 1$ RAM chips are needed to provide a memory capacity of $256K$ bytes?

A. 8
B. 32
C. 64
D. 128

gatecse-2009 digital-logic memory-interfacing easy isro2015

Answer key☟

8.21.2 Memory Interfacing: ISRO CSE 2014 | Question: 25

Suppose you want to build a memory with 4 byte words and a capacity of 2^{21} bits. What is type of decoder required if the memory is built using $2K \times 8$ RAM chips?

A. 5 to 32 B. 6 to 64 C. 4 to 16 D. 7 to 128

digital-logic memory-interfacing isro2014

Answer key

8.22 Min No Gates (2)

8.22.1 Min No Gates: ISRO CSE 2016 | Question: 7

The minimum number of NAND gates required to implement the Boolean function $A + A\overline{B} + A\overline{B}C$ is equal to

A. 0 (Zero) B. 1 C. 4 D. 7

digital-logic min-no-gates isro2016

Answer key

8.22.2 Min No Gates: ISRO CSE 2017 | Question: 23

What is the minimum number of two-input NAND gates used to perform the function of two-input OR gate?

A. One B. Two C. Three D. Four

isro2017 digital-logic min-no-gates

Answer key

8.23 Multiplexer (4)

8.23.1 Multiplexer: ISRO CSE 2008 | Question: 22

How many 2-input multiplexers are required to construct a 2^{10}-input multiplexer?

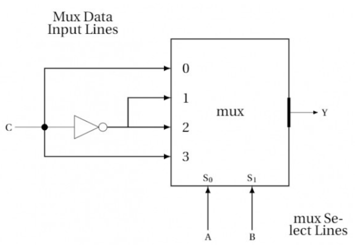
A. 1023 B. 31 C. 10 D. 127

isro2008 digital-logic multiplexer

Answer key

8.23.2 Multiplexer: ISRO CSE 2020 | Question: 10

Following Multiplexer circuit is equivalent to



A. Sum equation of full adder
B. Carry equation of full adder
C. Borrow equation for full subtractor
D. Difference equation of a full subtractor

isro-2020 digital-logic combinational-circuit multiplexer normal

Answer key☟

8.23.3 Multiplexer: ISRO CSE 2023 | Question: 70

Which of the following definitions is true

i. A digital multiplexer takes one input from many inputs and outputs the selected one
ii. Four elect lines are required for 1-16 multiplexer
iii. Eight OR gates are required to implement an octal to binary encoder

A. (i) and (ii) B. (ii) and (iii) C. (i) alone D. none of the above

isro-cse-2023 digital-logic digital-circuits multiplexer

Answer key☟

8.23.4 Multiplexer: ISRO2015-8

Minimum number of 2×1 multiplexers required to realize the following function, $f = \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}\,\overline{C}$

Assume that inputs are available only in true form and Boolean a constant 1 and 0 are available.

A. 1 B. 2 C. 3 D. 7

isro2015 digital-logic multiplexer

Answer key☟

8.24 Number Representation (25)

8.24.1 Number Representation: GATE CSE 1995 | Question: 2.12, ISRO2015-9

The number of 1's in the binary representation of $(3*4096 + 15*256 + 5*16 + 3)$ are:

A. 8 B. 9 C. 10 D. 12

gate1995 digital-logic number-representation normal isro2015

Answer key☟

8.24.2 Number Representation: GATE CSE 2005 | Question: 16, ISRO2009-18, ISRO2015-2

The range of integers that can be represented by an n bit $2's$ complement number system is:

A. $-2^{n-1} \text{ to } (2^{n-1}-1)$ B. $-(2^{n-1}-1) \text{ to } (2^{n-1}-1)$
C. $-2^{n-1} \text{ to } 2^{n-1}$ D. $-(2^{n-1}+1) \text{ to } (2^{n-1}-1)$

gatecse-2005 digital-logic number-representation easy isro2009 isro2015

Answer key☟

8.24.3 Number Representation: GATE CSE 2009 | Question: 5, ISRO2017-57

$(1217)_8$ is equivalent to

A. $(1217)_{16}$ B. $(028F)_{16}$ C. $(2297)_{10}$ D. $(0B17)_{16}$

gatecse-2009 digital-logic number-representation isro2017

Answer key☟

8.24.4 Number Representation: GATE IT 2006 | Question: 7, ISRO2009-41

The addition of $4-bit$, two's complement, binary numbers 1101 and 0100 results in

A. 0001 and an overflow
B. 1001 and no overflow
C. 0001 and no overflow
D. 1001 and an overflow

gateit-2006 digital-logic number-representation normal isro2009

Answer key☟

8.24.5 Number Representation: ISRO CSE 2007 | Question: 04

When two numbers are added in excess-3 code and the sum is less than 9, then in order to get the correct answer it is necessary to

A. subtract 0011 from the sum
B. add 0011 to the sum
C. subtract 0110 from the sum
D. add 0110 to the sum

isro2007 digital-logic number-representation

Answer key☟

8.24.6 Number Representation: ISRO CSE 2007 | Question: 18

The number of digit 1 present in the binary representation of
$3 \times 512 + 7 \times 64 + 5 \times 8 + 3$ is

A. 8
B. 9
C. 10
D. 12

isro2007 digital-logic number-representation

Answer key☟

8.24.7 Number Representation: ISRO CSE 2007 | Question: 38

The Hexadecimal equivalent of 01111100110111100011 is

A. CD73E
B. ABD3F
C. 7CDE3
D. FA4CD

isro2007 digital-logic number-representation

Answer key☟

8.24.8 Number Representation: ISRO CSE 2007 | Question: 49

One approach to handling fuzzy logic data might be to design a computer using ternary (base-3) logic so that data could be stored as “true,” “false,” and “unknown.” If each ternary logic element is called a flit, how many flits are required to represent at least 256 different values?

A. 4
B. 5
C. 6
D. 7

isro2007 digital-logic number-representation

Answer key☟

8.24.9 Number Representation: ISRO CSE 2008 | Question: 13

If $N^2 = (7601)_8$ where N is a positive integer, then the value of N is

A. $(241)_5$
B. $(143)_6$
C. $(165)_7$
D. $(39)_{16}$

isro2008 digital-logic number-representation

Answer key☟

8.24.10 Number Representation: ISRO CSE 2008 | Question: 20

If $(12x)_3 = (123)_x$, then the value of x is

A. 3 B. 3 or 4 C. 2 D. None of these

isro2008 digital-logic number-representation

Answer key☟

8.24.11 Number Representation: ISRO CSE 2009 | Question: 39 , GATE2008-IT_15

A processor that has carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two $2's$ complement numbers 01001101 and 11101001. After the execution of this addition operation, the status of the carry, overflow and sign flags, respectively will be

A. $1, 1, 0$ B. $1, 0, 0$ C. $0, 1, 0$ D. $1, 0, 1$

isro2009 number-representation digital-logic

Answer key☟

8.24.12 Number Representation: ISRO CSE 2014 | Question: 27

How many different BCD numbers can be stored in 12 switches ? (Assume two position or on-off switches).

A. 2^{12} B. $2^{12} - 1$ C. 10^{12} D. 10^3

isro2014 digital-logic number-representation

Answer key☟

8.24.13 Number Representation: ISRO CSE 2016 | Question: 14

If $12A7C_{16} = X_8$ then the value of X is

A. 224174 B. 425174 C. 6173 D. 225174

digital-logic number-representation isro2016

Answer key☟

8.24.14 Number Representation: ISRO CSE 2016 | Question: 17

Which of the following binary number is the same as its $2's$ complement ?

A. 1010 B. 0101 C. 1000 D. 1001

digital-logic number-representation isro2016

Answer key☟

8.24.15 Number Representation: ISRO CSE 2018 | Question: 19

Given $\sqrt{224_r} = 13_r$ the value of radix r is

a. 10 b. 8 c. 6 d. 5

isro2018 number-representation digital-logic

Answer key☟

8.24.16 Number Representation: ISRO CSE 2018 | Question: 3

If a variable can take only integral values from 0 to n, where n is an integer, then the variable can be represented as a bit-field whose width is (the log in the answer are to the base 2, and $\lceil \log n \rceil$ means the floor of $\log n$)

a. $\lceil \log(n) \rceil + 1$ bits
b. $\lceil \log(n-1) \rceil + 1$ bits
c. $\lceil \log(n+1) \rceil + 1$ bits
d. None of the above

isro2018 number-representation digital-logic

Answer key

8.24.17 Number Representation: ISRO CSE 2018 | Question: 32

A computer uses ternary system instead of the traditional systen, An n bit string in the binary system will occupy

a. $3+n$ ternary digits
b. $2n/3$ ternary digits
c. $n \log_2 3$ ternary digits
d. $n \log_3 2$ ternary digits

isro2018 digital-logic number-representation

Answer key

8.24.18 Number Representation: ISRO-2013-13

The number 1102 in base 3 is equivalent to 123 in which base system?

A. 4 B. 5 C. 6 D. 8

isro2013 number-representation

Answer key

8.24.19 Number Representation: ISRO-2013-17

Two eight bit bytes 11000011 and 01001100 are added. What are the values of the overflow, carry and zero flags respectively, if the arithmetic unit of the CPU uses 2's complement form?

A. $0, 1, 1'$ B. $1, 1, 0$ C. $1, 0, 1$ D. $0, 1, 0$

isro2013 digital-logic number-representation

Answer key

8.24.20 Number Representation: ISRO-2013-77

The binary equivalent of the decimal number 42.75 is

A. 101010.110 B. 100110.101 C. 101010.101 D. 100110.110

isro2013 digital-logic number-representation

Answer key

8.24.21 Number Representation: ISRO-2013-8

Which logic gate is used to detect overflow in 2's compliment arithmetic?

A. OR gate B. AND gate C. NAND gate D. XOR gate isro2013 number-representation

Answer key

8.24.22 Number Representation: ISRO-DEC2017-39

A computer with $32\text{-}bit$ word size uses $2's$ complement to represent numbers, The range of integers
that can be represented by this computer is

A. -2^{32} to 2^{32}
B. -2^{31} to $2^{32}-1$
C. -2^{31} to $2^{31}-1$
D. -2^{31} to $2^{32}-1$

isrodec2017 number-representation co-and-architecture

Answer key

8.24.23 Number Representation: ISRO-DEC2017-40

Let $M = 11111010$ and $N = 00001010$ be two $8\text{-}bit$ two's complement number.Their product in two's complement is

A. 11000100 B. 10011100 C. 10100101 D. 11010101

isrodec2017 number-representation digital-logic

Answer key

8.24.24 Number Representation: ISRO2015-6

The code which uses 7 bits to represent a character is :

A. ASCII B. BCD C. EBCDIC D. Gray

isro2015 digital-logic number-representation

Answer key

8.24.25 Number Representation: ISRO2015-78

The decimal number has 64 digits. The number of bits needed for its equivalent binary representation is?

A. 200 B. 213 C. 246 D. 277

isro2015 digital-logic number-representation

Answer key

8.25 Pla (1)

8.25.1 Pla: ISRO-2013-29

How many programmable fuses are required in a PLA which takes 16 inputs and gives 8 outputs? It has to use 8 OR gates and 32 AND gates.

A. 1032 B. 776 C. 1284 D. 1536

isro2013 digital-logic pla

Answer key

8.26 Shift Registers (1)

8.26.1 Shift Registers: ISRO CSE 2023 | Question: 71

The time delay obtained through an 8-bit serial register with 400 MHz clock is:

a. 20 ns
b. 2.5 µs
c. 20 µs
d. 2.5 µs

isro-cse-2023 shift-registers digital-circuits

Answer key☟

Answer Keys

8.0.1	B	8.0.2	D	8.0.3	C	8.0.4	X	8.0.5	C
8.0.6	D	8.0.7	D	8.1.1	B	8.1.2	C	8.1.3	D
8.1.4	D	8.1.5	B	8.1.6	C	8.2.1	A	8.3.1	C
8.3.2	A	8.3.3	A	8.3.4	X	8.3.5	B	8.3.6	B
8.3.7	B	8.3.8	C	8.3.9	C	8.3.10	A	8.3.11	A
8.3.12	B	8.3.13	A	8.4.1	B	8.5.1	B	8.5.2	C
8.6.1	B	8.6.2	C	8.7.1	A	8.7.2	X	8.7.3	A
8.7.4	C	8.7.5	C	8.7.6	D	8.7.7	A	8.7.8	B
8.7.9	A	8.7.10	A	8.7.11	A	8.7.12	B	8.7.13	A
8.7.14	B	8.8.1	A	8.9.1	C	8.10.1	X	8.10.2	X
8.11.1	C	8.11.2	B	8.12.1	C	8.12.2	D	8.12.3	A
8.12.4	C	8.12.5	A	8.12.6	B	8.13.1	A	8.13.2	D
8.13.3	C	8.13.4	C	8.13.5	D	8.13.6	C	8.14.1	C
8.14.2	B	8.15.1	C	8.16.1	C	8.17.1	B	8.18.1	A
8.19.1	D	8.20.1	D	8.21.1	C	8.21.2	A	8.22.1	A
8.22.2	C	8.23.1	A	8.23.2	A;D	8.23.3	A	8.23.4	B
8.24.1	C	8.24.2	A	8.24.3	B	8.24.4	C	8.24.5	A
8.24.6	B	8.24.7	C	8.24.8	C	8.24.9	B	8.24.10	D
8.24.11	B	8.24.12	D	8.24.13	D	8.24.14	C	8.24.15	D
8.24.16	A	8.24.17	D	8.24.18	B	8.24.19	D	8.24.20	A
8.24.21	D	8.24.22	C	8.24.23	A	8.24.24	A	8.24.25	B
8.25.1	C	8.26.1	A						